

1. Multimodal Observation

RGB Image



3D Geometry



Language Instruction

"Pick up the wooden chair and move it to the left."



2. Object-centric Representation

Shape



Semantic



Physical Cues



Size / Mass



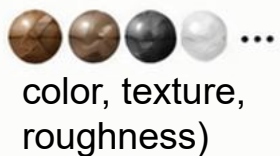
Material Hint



Physical Context

3. Physical Understanding

Appearance



Material & Mass



(density, stiffness, mass)

Dynamics



(friction, deformation, damping)

Physically Grounded Properties

4. Manipulation Affordance

Grasp



Push



Pull



Lift/Place



...

Actionable Affordance Map & Modes

5. Actionable Object Representation



Geometry



Semantics



Physical Properties



Affordance Modes

Robot Manipulation

